

CÉSAR A. URIBE

+1 (217) 721 6156, cauribe@rice.edu
<https://cauribe.rice.edu>

RESEARCH INTERESTS: • optimization theory • network science • statistical learning • belief systems • optimal transport

APPOINTMENTS

Rice University	Houston, TX
Louis Owen Assistant Professor, Department of Electrical and Computer Engineering	2021–
Rice Academy Senior Fellow	2022–
Ken Kennedy Institute Member	2022–
National Library of Medicine (NLM) Training Program Faculty	2022–
KTH Royal Institute of Technology	Stockholm, Sweden
Digital Futures 2024 Early Career Fellow	2024 (Summer)
Massachusetts Institute of Technology	Cambridge, MA
Post Doctoral Associate, Laboratory for Information and Decision Systems (LIDS),	2018-2020 (Spring)
Post Doctoral Associate, MITx MicroMasters program in Statistics and Data Science,	2020 (Summer/Fall)
Host: A. Jadbabaie	
Moscow Institute of Physics, and Technology	Moscow, Russian Federation
Visiting Assistant Professor, Phystech School of Applied Mathematics and Informatics,	2019-2022

EDUCATION

University of Illinois at Urbana-Champaign	Urbana, IL
Ph.D. in Electrical and Computer Engineering (GPA 3.90/4)	2013-2018
Thesis: <i>Efficient and Scalable Algorithms for Distributed Learning, Optimization, and Belief Systems over Networks</i>	
Committee: A. Nedić, A. Olshevsky, M. Raginsky, R. Srikant, T. Başar (Chair)	
M.Sc. in Applied Mathematics: Optimization and Algorithms	2016
Delft University of Technology	Delft, Netherlands
M.Sc. in Systems and Control (<i>Cum Laude</i> , GPA 8.7/10)	2011-2013
Thesis: <i>Control of Finite Stochastic Systems with non-Classical Information Patterns</i>	
Advisers: J. van Schuppen and T. Keviczky	
Universidad de Antioquia	Medellin, Colombia
B.Sc. in Electronics Engineering (<i>with Honors</i> , GPA 4.28/5)	2005-2010

SELECTED PUBLICATIONS (★ denotes alphabetical ordering) [Google Scholar](#)

Journal Papers

- [100] T. Qin, S. R. Etesami, and CAU, “Faster convergence of local SGD for over-parameterized models,” *Transactions on Machine Learning Research*, 2024
- [99] S. Gracy, J. I. Caiza, P. Paré, and CAU, “Multi-competitive time-varying networked SIS model with an infrastructure network,” *IFAC Journal of Systems and Control*, p. 100254, 2024
- [98] S. Gracy, M. Ye, B. Anderson, and CAU, “Towards understanding the endemic behavior of a competitive tri-virus sis networked model,” *SIAM Journal on Applied Dynamical Systems*, vol. 23, no. 2, pp. 1372–1410, 2024
- [97] P. Dvurechensky, P. Ostroukhov, A. Gasnikov, CAU, and A. Ivanova, “Near-optimal tensor methods for minimizing the gradient norm of convex functions and accelerated primal–dual tensor methods,” *Optimization Methods and Software*, vol. 0, no. 0, pp. 1–36, 2024
- [96] M. T. Toghani, S. Lee, and CAU, “PersA-FL: personalized asynchronous federated learning,” *Optimization Methods and Software*, vol. 0, no. 0, pp. 1–38, 2024
- [95] T. Qin, S. R. Etesami, and CAU, “The role of local steps in local SGD,” *Optimization Methods and Software*, vol. 0, no. 0, pp. 1–27, 2023
- [94] J. Castellanos, C. A. Correa-Floreza, A. Garcés, G. Ordóñez-Platad, CAU, and D. Patino, “An energy management system model with power quality constraints for unbalanced multi-microgrids interacting in a local energy market,” *Applied Energy*, vol. 343, p. 121149, 2023
- [93] M. T. Toghani, S. Lee, and CAU, “Pars-push: Personalized, asynchronous and robust decentralized optimization,” *IEEE Control Systems Letters*, vol. 7, pp. 361–366, 2023
- [92] J. L. Kim, M. T. Toghani, CAU, and A. Kyriillidis, “Local stochastic factored gradient descent for distributed quantum state tomography,” *IEEE Control Systems Letters*, vol. 7, pp. 199–204, 2023
- [91] P. Dvurechensky, D. Kamzolov, A. Lukashevich, S. Lee, E. Ordentlich, CAU, and A. Gasnikov, “Hyperfast second-order local solvers for efficient statistically preconditioned distributed optimization,” *EURO Journal on Computational Optimization*, vol. 10, p. 100045, 2022

- [90] M. T. Toghiani and CAU, “On arbitrary compression for decentralized consensus and stochastic optimization over directed networks,” *European Journal of Control*, vol. 68, p. 100682, 2022, 2022 European Control Conference Special Issue
- [89] B. Turan, CAU, H.-T. Wai, and M. Alizadeh, “Robust distributed optimization with randomly corrupted gradients,” *IEEE Transactions on Signal Processing*, vol. 70, pp. 3484–3498, 2022
- [88] F. Arqu , CAU, and C. Ocampo-Martinez, “Approximate Wasserstein attraction flows for dynamic mass transport over networks,” *Automatica*, vol. 143, p. 110432, 2022. **September 2022 Editor’s Choice**
- [87] M. T. Toghiani and CAU, “Communication-efficient distributed cooperative learning with compressed beliefs,” *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1215–1226, 2022
- [86] CAU, A. Olshevsky, and A. Nedi , “Nonasymptotic concentration rates in cooperative learning—Part I: Variational non-bayesian social learning,” *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1128–1140, 2022
- [85] —, “Nonasymptotic concentration rates in cooperative learning—Part II: Inference on compact hypothesis sets,” *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1141–1153, 2022
- [84] J. Z. Hare, CAU, L. Kaplan, and A. Jadbabaie, “A general framework for distributed inference with uncertain models,” *IEEE Transactions on Signal and Information Processing over Networks, Presented at ICASSP 2022*, vol. 7, pp. 392–405, 2021
- [83] D. E. Ochoa, J. I. Poveda, CAU, and N. Quijano, “Robust optimization over networks using distributed restarting of accelerated dynamics,” *IEEE Control Systems Letters*, vol. 5, no. 1, pp. 301–306, 2021
- [82] B. Turan, CAU, H.-T. Wai, and M. Alizadeh, “Resilient primal–dual optimization algorithms for distributed resource allocation,” *IEEE Transactions on Control of Network Systems*, vol. 8, no. 1, pp. 282–294, 2021
- [81] CAU, S. Lee, A. Gasnikov, and A. Nedi , “A dual approach for optimal algorithms in distributed optimization over networks,” *Optimization Methods and Software*, vol. 36, no. 1, pp. 171–210, 2021
- [80] J. Z. Hare, CAU, L. M. Kaplan, and A. Jadbabaie, “Non-Bayesian social learning with uncertain models,” *IEEE Transactions on Signal Processing*, vol. 68, pp. 4178–4193, 2020
- [79] H.-T. Wai, W. Shi, CAU, A. Nedi , and A. Scaglione, “Accelerating incremental gradient optimization with curvature information,” *Computational Optimization and Applications*, vol. 76, no. 2, pp. 347–380, Jun 2020
- [78] A. Rogozin, CAU, A. V. Gasnikov, N. Malkovsky, and A. Nedi , “Optimal distributed convex optimization on slowly time-varying graphs,” *IEEE Transactions on Control of Network Systems*, vol. 7, no. 2, pp. 829–841, 2020
- [77] ★ A. Nedi , A. Olshevsky, and CAU, “Graph-theoretic analysis of belief system dynamics under logic constraints,” *Scientific Reports*, vol. 9, no. 1, p. 8843, 2019
- [76] ★ —, “Fast Convergence Rates for Distributed Non-Bayesian Learning,” *IEEE Transactions on Automatic Control*, vol. 62, no. 11, pp. 5538–5553, Nov. 2017
- [75] CAU and C. Isaza, “Expert knowledge-guided feature selection for data-based industrial process monitoring,” *Revista Facultad de Ingenier a Universidad de Antioquia*, no. 65, pp. 112–125, 12 2012

Preprints and Submitted

- [74] S. Gracy, B. D. Anderson, M. Ye, and CAU, “Networked Competitive Bivirus SIS spread with Higher Order Interactions,” *Submitted to SIAM Optimization and Control*, <https://hal.science/hal-04669193>, 2024
- [73] M. Guerrero, C. S nchez-Giraldo, CAU, V. Mart nez, and C. Isaza, “Graphical representation of landscape heterogeneity identification through unsupervised acoustic analysis,” *Submitted to Methods in Ecology and Evolution*, 2024
- [72] D. Pasechnyuk, P. Dvurechensky, CAU, and A. Gasnikov, “Decentralized convex optimisation with probability-proportional-to-size quantization,” June 2024, manuscript submitted to NeurIPS 2024
- [71] ★ X. Jiang, E. D. H. Nguyen, CAU, and B. Ying, “Sparse factorization of the square all-ones matrix of arbitrary order,” *submitted SIAM Journal Matrix Analysis and Applications*, 2024
- [70] M. E. Lugo-V lez, A. R. Guhlincozzi, S. Gupta, and CAU, “Goya boycott: A protest in code/space,” *Submitted to Social & Cultural Geography*, 2024
- [69] B. Wu and CAU, “Frequentist guarantees of distributed (non)-Bayesian inference,” *arXiv preprint arXiv:2311.08214 Submitted to JMLR*, 2023
- [68] E. D. H. Nguyen, X. Jiang, B. Ying, and CAU, “On graphs with finite-time consensus and their use in gradient tracking,” *arXiv:2311.01317, submitted SIAM Journal on Optimization*, 2023
- [67] R. Krawtschenko, CAU, A. Gasnikov, and P. Dvurechensky, “Distributed optimization with quantization for computing Wasserstein barycenters,” *arXiv:2010.14325*, 2022
- [66] CAU and A. Jadbabaie, “A distributed cubic-regularized newton method for smooth convex optimization over networks,” *arXiv preprint arXiv:2007.03562*, 2020
- [65] ★ P. Dvurechensky, M. Staudigl, and CAU, “Generalized self-concordant Hessian-barrier algorithms,” *arXiv:1911.01522*, 2019
- [64] ★ A. Nedi , A. Olshevsky, and CAU, “Distributed learning for cooperative inference,” *arXiv:1704.02718*, 2017

Machine Learning Conference Papers

- [63] T. M. Nguyen, CAU, T. M. Nguyen, and R. Baraniuk, “PIDformer: transformer meets control theory,” *Accepted Forty-first International Conference on Machine Learning ICML 2024*, 2024

- [62] J. L. Kim, M. T. Toghani, **CAU**, and A. Kyrillidis, “Adaptive federated learning with auto-tuned clients,” *Accepted to ICLR 2024, arXiv preprint arXiv:2306.11201*, 2023
- [61] A. Gasnikov, P. Dvurechensky, E. Gorbunov, E. Vorontsova, D. Selikhanovych, and **CAU**, “Optimal tensor methods in smooth convex and uniformly convex optimization,” in *Proceedings of the Thirty-Second Conference on Learning Theory*, vol. 99, 2019, pp. 1374–1391
- [60] A. Gasnikov, P. Dvurechensky, E. Gorbunov, E. Vorontsova, D. Selikhanovych, **CAU**, B. Jiang, H. Wang, S. Zhang, S. Bubeck, Q. Jiang, Y. T. Lee, Y. Li, and A. Sidford, “Near optimal methods for minimizing convex functions with lipschitz p -th derivatives,” in *Proceedings of the Thirty-Second Conference on Learning Theory*, ser. Proceedings of Machine Learning Research, vol. 99. PMLR, 25–28 Jun 2019, pp. 1392–1393
- [59] A. Kroshnin, N. Tupitsa, D. Dvinskikh, P. Dvurechensky, A. Gasnikov, and **CAU**, “On the complexity of approximating Wasserstein barycenters,” in *Proceedings of the 36th International Conference on Machine Learning*, vol. 97, 2019, pp. 3530–3540
- [58] P. Dvurechenskii, D. Dvinskikh, A. Gasnikov, **CAU**, and A. Nedić, “Decentralize and randomize: Faster algorithm for Wasserstein barycenters,” in *Advances in Neural Information Processing Systems 31*, 2018, pp. 10 760–10 770 **Spotlight Presentation, Top 4%**.

Control Theory Conference Papers

- [57] A. Aravind, M. T. Toghani, and **CAU**, “A Moreau envelope approach for LQR meta-policy estimation,” *arXiv 2403.17364, Accepted to CDC2024*, 2024
- [56] A. G. Zalles, K. M. Hung, A. E. Finneran, L. Beaudrot, and **CAU**, “An optimal transport approach for network regression,” *arXiv preprint arXiv:2406.12204 Accepted IEEE Conference on Control Technology and Applications (CCTA2024)*, 2024
- [55] M. F. Arevalo-Castiblanco, Y. Wi, M. Cescon, and **CAU**, “An application of model reference adaptive control for multi-agent synchronization in drone networks,” *Accepted IEEE Conference on Control Technology and Applications (CCTA2024)*, 2024
- [54] C. A. Taveras, S. Segarra, and **CAU**, “Efficient path planning with soft homology constraints,” *Accepted IEEE Conference on Control Technology and Applications (CCTA2024)*, 2024
- [53] S. Gracy, J. Liu, T. Basar, and **CAU**, “A discrete-time networked competitive bivirus SIS model,” in *2024 European Control Conference (ECC)*, 2024, pp. 3398–3403
- [52] I. Lau, S. Ma, and **CAU**, “Decentralized and equitable optimal transport,” *Accepted to American Control Conference 2024*, 2024
- [51] S. Gracy, B. Anderson, M. Ye, and **CAU**, “Competitive networked bivirus SIS spread over hypergraphs,” *Accepted American Control Conference 2024, arXiv preprint arXiv:2309.14230*, 2024
- [50] M. T. Toghani, S. Perez-Salazar, and **CAU**, “On first-order meta-reinforcement learning with Moreau envelopes,” in *2023 62nd IEEE Conference on Decision and Control (CDC)*, 2023, pp. 4176–4181
- [49] E. Nguyen, S. Alghunaim, K. Yuan, and **CAU**, “On the performance of gradient tracking with local updates,” in *2023 62nd IEEE Conference on Decision and Control (CDC)*, 2023, pp. 4309–4313
- [48] S. Gracy, Y. Wang, **CAU**, and P. Paré, “Multi-competitive virus spread over a time-varying networked SIS model with an infrastructure network,” *IFAC-PapersOnLine*, vol. 56, no. 2, pp. 19–24, 2023, 22nd IFAC World Congress
- [47] S. Gracy, M. Ye, B. Anderson, and **CAU**, “On the endemic behavior of a competitive tri-Virus SIS networked model,” in *2023 American Control Conference*, 2023, pp. 2313–2318
- [46] Y. Wang, S. Gracy, **CAU**, H. Ishii, and K. H. Johansson, “A state feedback controller for mitigation of continuous-time networked SIS epidemics,” *IFAC-PapersOnLine*, vol. 55, no. 41, pp. 89–94, 2022, 4th IFAC Workshop on Cyber-Physical and Human Systems CPHS 2022
- [45] M. T. Toghani, S. Lee, and **CAU**, “Unbounded gradients in federated learning with buffered asynchronous aggregation,” in *2022 58th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, 2022, pp. 1–8
- [44] T. Qin, S. R. Etesami, and **CAU**, “Decentralized federated learning for over-parameterized models,” in *2022 IEEE 61st Conference on Decision and Control*, 2022, pp. 5200–5205
- [43] T. Anh-Nguyen and **CAU**, “On distributed exact sparse linear regression over networks,” in *2022 IEEE 61st Conference on Decision and Control*, 2022, pp. 6491–6496
- [42] S. M. Farrell, A. Veeraraghavan, A. Sabharwal, and **CAU**, “Distributed generalized Wirtinger flow for interferometric imaging on networks,” *IFAC-PapersOnLine*, vol. 55, no. 13, pp. 258–263, 2022, 9th IFAC Conference on Networked Systems NECSYS 2022
- [41] N. Mehrotra, A. Sabharwal, and **CAU**, “Consensus ADMM-based distributed simultaneous imaging & communication,” *IFAC-PapersOnLine*, vol. 55, no. 13, pp. 31–36, 2022, 9th IFAC Conference on Networked Systems NECSYS 2022
- [40] E. Trimbach, E. Nguyen, and **CAU**, “On acceleration of gradient-based empirical risk minimization using local polynomial regression,” in *2022 European Control Conference*, 2022, pp. 429–434
- [39] M. T. Toghani and **CAU**, “Scalable average consensus with compressed communications,” in *2022 American Control Conference*, 2022, pp. 3412–3417
- [38] D. E. Ochoa, J. I. Poveda, and **CAU**, “Computation-aware distributed optimization over networks: A hybrid dynamical systems approach,” in *Proceedings of the Workshop on Computation-Aware Algorithmic Design for Cyber-Physical Systems*. Association for Computing Machinery, 2021, p. 18–19

- [37] F. Arqu , **CAU**, and C. Ocampo-Martinez, “Application of wasserstein attraction flows for optimal transport in network systems,” in *2021 60th IEEE Conference on Decision and Control (CDC)*, 2021, pp. 4058–4063
- [36] B. Turan, **CAU**, H.-T. Wai, and M. Alizadeh, “On robustness of the normalized random block coordinate method for non-convex optimization,” in *2021 60th IEEE Conference on Decision and Control (CDC)*, 2021, pp. 974–980
- [35] M. Arevalo-Castiblanco, **CAU**, and E. Mojica-Nava, “Model reference adaptive control for online policy adaptation and network synchronization,” in *2021 60th IEEE Conference on Decision and Control*, 2021, pp. 4071–4076
- [34] J. Z. Hare, **CAU**, L. Kaplan, and A. Jadbabaie, “Toward active sequential hypothesis testing with uncertain models,” in *2021 60th IEEE Conference on Decision and Control*, 2021, pp. 3709–3716
- [33] T. Qin, S. R. Etesami, and **CAU**, “Communication-efficient decentralized local SGD over undirected networks,” in *2021 60th IEEE Conference on Decision and Control*, 2021, pp. 3361–3366
- [32] B. Turan, **CAU**, H.-T. Wai, and M. Alizadeh, “On robustness of the normalized subgradient method with randomly corrupted subgradients,” in *2021 American Control Conference*, 2021, pp. 965–971
- [31] E. Mojica-Nava, D. Yanguas-Rojas, and **CAU**, “Robust asynchronous and network-independent cooperative learning,” in *2021 American Control Conference*, 2021, pp. 1619–1624
- [30] N. Tupitsa, P. Dvurechensky, A. Gasnikov, and **CAU**, “Multimarginal optimal transport by accelerated alternating minimization,” in *2020 59th IEEE Conference on Decision and Control*, 2020, pp. 6132–6137
- [29] J. Z. Hare, **CAU**, L. Kaplan, and A. Jadbabaie, “Non-bayesian social learning with gaussian uncertain models,” in *2020 American Control Conference*, 2020, pp. 4484–4490
- [28] **CAU**, H. Wai, and M. Alizadeh, “Resilient distributed optimization algorithms for resource allocation,” in *IEEE Conference on Decision and Control*, 2019, pp. 8341–8346
- [27] **CAU**, J. Hare, L. Kaplan, and A. Jadbabaie, “Non-Bayesian social learning with uncertain models over time-varying directed graphs,” in *IEEE Conference on Decision and Control*, 2019, pp. 3635–3640
- [26] D. Ochoa, J. Poveda, **CAU**, and N. Quijano, “Hybrid robust optimal resource allocation with momentum,” in *IEEE Control and Decision Conference*, 2019, pp. 3954–3959
- [25] D. Dvinskikh, E. Gorbunov, A. Gasnikov, P. Dvurechensky, and **CAU**, “On primal and dual approaches for distributed stochastic convex optimization over networks,” in *IEEE Conference on Decision and Control*, 2019, pp. 7435–7440
- [24] J. Zhang, **CAU**, A. Mokhtari, and A. Jadbabaie, “Achieving acceleration in distributed optimization via direct discretization of the Heavy-Ball ODE,” in *Americal Control Conference*, 2019, pp. 3408–3413
- [23] **CAU** and A. Jadbabaie, “On increasing self-confidence in non-Bayesian social learning over time-varying directed graphs,” in *American Control Conference*, 2019, pp. 3532–3537
- [22] E. Baron-Prada, **CAU**, and E. Mojica-Nava, “A method for distributed transactive control in power systems based on the projected consensus algorithm,” *IFAC-PapersOnLine*, vol. 51, no. 23, pp. 379 – 384, 2018, 7th IFAC Workshop on Distributed Estimation and Control in Networked Systems NECSYS 2018
- [21] **CAU**, D. Dvinskikh, P. Dvurechensky, A. Gasnikov, and A. Nedi , “Distributed computation of Wasserstein barycenters over networks,” in *IEEE Conference on Decision and Control*, 2018, pp. 6544–6549
- [20] * A. Nedi , A. Olshevsky, W. Shi, and **CAU**, “Geometrically convergent distributed optimization with uncoordinated step-sizes,” in *American Control Conference*, 2017, pp. 3950–3955
- [19] * A. Nedi , A. Olshevsky, and **CAU**, “Distributed learning with infinitely many hypotheses,” in *IEEE Conference on Decision and Control*, 2016, pp. 6321–6326
- [18] * —, “A tutorial on distributed (non-Bayesian) learning: Problem, algorithms and results,” in *IEEE Conference on Decision and Control*, 2016, pp. 6795–6801
- [17] * —, “Network independent rates in distributed learning,” in *American Control Conference*, 2016, pp. 1072–1077
- [16] * —, “Nonasymptotic convergence rates for cooperative learning over time-varying directed graphs,” in *American Control Conference*, 2015, pp. 5884–5889
- [15] **CAU**, T. Keviczky, and J. H. van Schuppen, “Computing optimal control laws for finite stochastic systems with non-classical information patterns,” in *American Control Conference*, 2014, pp. 5742–5747
- [14] **CAU** and J. H. van Schuppen, “Analysis of signaling in a finite stochastic system motivated by decentralized control,” in *IEEE Conference on Decision and Control*, 2013, pp. 5884–5889

Signal Processing Conference Papers

- [13] **CAU**, J. Hare, L. Kaplan, and A. Jadbabaie, “On the inconsistency of estimation with uncertain models,” in *Accepted to Asilomar Conference 2024*, 2024
- [12] M. Zafari, D. Pandey, R. Doost-Mohammady, and **CAU**, “ADMM for Downlink Beamforming in Cell-Free Massive MIMO Systems,” in *Accepted to Asilomar 2024*, 2024
- [11] D. Pirhayatifard, M. T. Toghani, G. Balakrishnan, and **CAU**, “Improving denoising diffusion probabilistic models via exploiting shared representations,” in *2023 57th Asilomar Conference on Signals, Systems, and Computers*, 2023, pp. 789–793
- [10] M. T. Toghani and **CAU**, “Communication-efficient and fault-tolerant social learning,” in *2021 55th Asilomar Conference on Signals, Systems, and Computers*, 2021, pp. 1037–1042

- [9] J. Z. Hare, **CAU**, L. Kaplan, and A. Jadbabaie, “Communication constrained learning with uncertain models,” in *International Conference on Acoustics, Speech, and Signal Processing, ICASSP*, 2020, pp. 8609–8613
- [8] —, “On malicious agents in non-Bayesian social learning with uncertain models,” in *International Conference on Information Fusion*, 2019, pp. 1–8
- [7] ★ A. Nedić, A. Olshevsky, and **CAU**, “Distributed Gaussian learning over time-varying directed graphs,” in *Asilomar Conference on Signals, Systems and Computers*, 2016, pp. 1710–1714
- [6] **CAU** and C. Isaza, “Unsupervised feature selection based on fuzzy partition optimization for industrial processes monitoring,” in *2011 IEEE International Conference on Computational Intelligence for Measurement Systems and Applications (CIMSAP) Proceedings*, 2011, pp. 1–5
- [5] **CAU**, C. Isaza, and J. F. Florez-Arango, “Qualitative-fuzzy decision support system for monitoring patients with cardiovascular risk,” in *2011 Eighth International Conference on Fuzzy Systems and Knowledge Discovery (FSKD)*, vol. 3, 2011, pp. 1621–1625
- [4] **CAU**, C. Isaza, O. Gualdron, C. Duran, and A. Carvajal, “A wrapper approach based on clustering for sensors selection of industrial monitoring systems,” in *2010 International Conference on Broadband, Wireless Computing, Communication and Applications*, 2010, pp. 482–487

Book Chapters

- [3] F. S. Stonyakin, D. Dvinskikh, P. Dvurechensky, A. Kroshnin, O. Kuznetsova, A. Agafonov, A. Gasnikov, A. Tyurin, **CAU**, D. Pasechnyuk, and S. Artamonov, “Gradient methods for problems with inexact model of the objective,” in *Mathematical Optimization Theory and Operations Research*. Springer International Publishing, 2019, pp. 97–114
- [2] **CAU** and J. H. van Schuppen, “Signaling of information,” in *Coordination Control of Distributed Systems*, J. H. van Schuppen and T. Villa, Eds. Springer International Publishing, 2015, pp. 165–172
- [1] C. Bedoya, **CAU**, and C. Isaza, “Unsupervised feature selection based on fuzzy clustering for fault detection of the tennessee eastman process,” in *Advances in Artificial Intelligence – IBERAMIA 2012*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2012, pp. 350–360

HONORS AND AWARDS

- **PI**: *Enabling Next-Generation Performance for Efficient Distributed Training*
Google Research Scholar Award, 2024
- **Co-PI** with Shiqian Ma (CMOR): *INFORMS Optimization Society Conference 2024*
Creative Ventures Funds, Rice University 2023 & **Office of Naval Research**, 2024
- **PI**: *The Next Generation Archive: Towards Systematic Qualitative Analysis of Social Media Enhancing the Humanities*
Digital Humanities Seed Funding, Ken Kennedy Institute 2023
- **PI** with Diego Patino (PUJ): *Rice University-Universidad Javeriana Research Exchange for the 4th Industrial Revolution*
US Department of State 100,000 Strong in the Americas Innovation Fund 2023
- **Co-PI** with Tasos Kyrillidis and Chris Jermaine: *Future-proof Distributed/Federated ML Computing*
Engineering Innovation Conference and Workshop Fund Award & Creative Ventures Funds 2023
- **PI**: *Building Capacities for Systematic and Qualitative Analysis in the Social Media Archive*
Fondren Fellows Program Rice University 2022
- **PI**: *Towards the Tropical Forest Connectome: Soundscape Ecology Network Science for Understanding of Conservation States*,
2022 Ralph E. Powe Junior Faculty Enhancement Award
- **Co-PI** with Yingyan Lin (Rice) and Anshumali Shrivastava (Rice): *SHF: DILSE: Codesigning Decentralized Incremental Learning System via Streaming Data Summarization on Edge*
National Science Foundation NSF 2022
- **Co-PI** with Lydia Beaudrot (Rice): *MSA: Quantifying the drivers of mammal food web networks over space and time in Sub-Saharan Africa*
National Science Foundation NSF 2022
- **Co-PI** with Lydia Beaudrot (Rice): *Identifying anthropogenic impacts on tropical wildlife through multi-scale and multi-modal network ecology using graph matching and alignment*
Creative Ventures Fund: Sustainable Futures Fund 2022
- **PI** with Sebastian Acosta (BCM): *Geometric Methods for Detection of Junctional Ectopic Tachycardia in Postoperative Children*
Creative Ventures Fund: ARPA-H Strategic Initiative Seed Fund 2021
- **PI**: *INFORMS en Español, INFORMS Diversity, Equity, and Inclusion* –Ambassadors Program Award
INFORMS Institute for Operations Research and the Management Sciences
- **Co-PI** with Alexander Gasnikov (MIPT): *Acceleration for Data Science and Machine Learning*
Yahoo! Research –Faculty and Research Engagement Program (FREP) 2019
- **PI** with Andrés Salazar (MIT): *Bringing MIT STEM Education to the East Boston Community*
MIT Committee on Race and Diversity –CRD Grant 2018

- *Career Champion Award, Class of 2024* Center for Career Development (CCD) Rice University, 2024.
- *Career Champion Award, Class of 2022* Center for Career Development (CCD) Rice University, 2022.
- *Best Poster Award* Midwest Machine Learning Symposium (MMLS), 2018.
- *Best Poster Presentation Award* 7th Midwest Workshop on Control and Game Theory, 2018.
- *Best Talk Award* Machine Learning Session, CSL Student Conference, 2018.
- *Best Presentation* Learning and Estimation in Networks Session, American Control Conference, 2016.
- *Outstanding Poster Award* Stochastic Networks Conference, 2016.
- *Best Talk Award* Security Session, CSL Student Conference, 2016.
- *Teacher Ranked as Excellent* (Student selected rank) University of Illinois at Urbana-Champaign, Fall 2016.
- *Best Presentation* Decentralized Control Session, American Control Conference, 2014.
- *Travel Award* ACC 2014, ICML 2019, COLT2019

TEACHING

- **Instructor:** Course (6 hours), Distributed Optimization, Texas State University, Spring 2024.
- **Instructor:** Course (10 hours), Introduction to AI, Genesys Works and ION-Rice University, Summer 2022.
- **Instructor:** ELEC 303 Random Signals, Rice University, Fall' 22, Fall' 23.
- **Lecturer:** Course (32 hours) on Selected Topics on ML, Pontificia Universidad Javeriana, Colombia. Summer 2022
- **Instructor: (NEW)** ELEC 570 Distributed Methods for Optimization and ML, Rice University, Fall 21, Spring 22, Spring 23.
- **Guest Lecturer:** COMP 566 AI Ethics, Rice University, Spring 2022.
- **Instructor:** ELEC 490 Independent Research for Undergraduates, Rice University, Fall 2021.
- **Recitation Instructor:** 6.431 Probability, MITx MicroMasters in Statistics and Data Science Fall 2020.
- **Module Developer:** 6.419 Statistics, Computation and Applications, MITx MicroMasters in Statistics and Data Science Spring 2020.
- **Teaching Assistant:** ECE534 Random Processes, UIUC-Fall 2016. *Teacher Ranked as Excellent by Students.*
- **Lecturer:** Course (8-hours) on Distributed Optimization & Learning, Pontificia Universidad Javeriana, Colombia. Fall 2019
- **Guest Lecturer:** Control of Discrete Stochastic Systems, TUDelft-Spring 2017

ADVOCACY, MENTORING & DEVELOPMENT

- **Research Group**
 - Carlos Taveras, **GEM Fellow**, co-advised with Santiago Segarra, PhD ECE Rice University.
 - * IFAC CPHS Fellow International Federation on Automatic Control 2022
 - * Ken Kennedy-HPE Cray Graduate Fellowship 2023
 - * Invited to the 2023 Konstanz School of Collective Behaviour
 - * IEEE Conference on Control Technology and Applications (CCTA2024) Travel Award 2024
 - * Interacting Particle Systems Workshop Travel Award ICERM, Brown University 2024
 - Felipe Arevalo, **Fulbright Visiting Scholar 2022 - 2024**, PhD ECE Universidad Nacional de Colombia.
 - * IEEE Control Systems Society Graduate Collaboration Fellowship 2023
 - * IFAC CPHS Fellow International Federation on Automatic Control 2022
 - * LatinX in AI Mentor at NeurIPS 2021 LXAI 2021
 - * LatinX in AI Chair at NeurIPS 2022 LXAI 2022
 - * LatinX in AI Chair at AISTATS 2023 LXAI 2023
 - Alex Zalles, **Undergraduate Researcher Google-Rice Program**, NSF REU, Summer 2023, OR Rice University.
 - * Poster Session - Best Overall Rice University, School of Engineering 2023
 - * Rice Undergraduate Scholars Program Award Rice University 2024
 - * Admitted to the MIT Summer Research Program MSRP 2024
 - * IEEE Conference on Control Technology and Applications (CCTA2024) Travel Award 2024
 - Gabriel Ong, Summer NSF REU Undergraduate Researcher CS Rice University. 2024
 - Yifan Zhang, **SER Scholar**, CS Rice University, Summer 2024
- **Research Group Alumni**
 - Mohammad Taha Toghani, PhD ECE Rice University. (Now at Cubist)
 - * Lodieska Stockbridge Vaughn Fellowship Rice University 2023
 - * Optimal Transport in Data Science Workshop Travel Award ICERM, Brown University 2023
 - * Neural Information Processing Systems Travel Award NeurIPS 2022
 - * SIAM Conference on Mathematics of Data Science Travel Award SIAM 2022
 - * International Conference on Machine Learning Travel Award ICML 2022
 - * Stochastic Networks Conference Travel Award Cornell University 2022
 - * Princeton Machine Learning Theory Summer School Travel Award Princeton University 2022
 - * American Control Conference (ACC) Travel Award Rice University 2022
 - * North American School of Information Theory Travel Award UCLA/NSF 2022
 - * Rice Engineering Alumni (REA) Graduate Student Travel Grant Rice University 2021-2022
 - * SIAM Texas-Louisiana Section (TXLA21) Travel Award SIAM 2021
 - * IEEE Conference on Decision and Control (CDC) Travel Award IEEE 2020-2021
 - * SIAM Conference on Optimization (OP21) Travel Award SIAM 2021

- * Society of Iranian-American Women for Education (SIAWE) Fellowship SIAWE 2020-2021
- Edward Duc Hien Nguyen, MSc ECE Rice University.
 - * National Library of Medicine Fellowship in Biomedical Informatics and Data Science National Library of Medicine 2021
 - * IEEE Conference on Decision and Control (CDC) Travel Award IEEE 2023
 - * Ken Kennedy-HPE Cray Graduate Fellowship 2023
- Sebin Jose Gracy, Postdoctoral Associate, Rice University.
 - * Rice Academy Fellowship
 - * Now Assistant Professor ECE Dept South Dakota School of Mines, Jan 2024
- Kai Hung, Undergraduate Researcher CS Rice University. 2022-2024 (Now PhD at MIT)
 - * NSF Graduate Research Fellowships Program (GRFP) 2024
 - * Rice Undergraduate Scholars Program Award Rice University 2023
 - * CMMRS Pre-Doctoral Research School Fellowship Cornell, Maryland, Max Planck Institute of Software Systems 2023
 - * Louise J. Walsh Scholarship in Engineering Rice University, George R. Brown School of Engineering 2023
- Paul Camargo, **Fulbright Visiting Scholar 2023 - 2024**, PhD Mathematics Universidad Nacional de Colombia.
- Maria Jose Guerrero, **Visiting Scholar Fall 2022/ Spring 2024**, Ph.D. Student Universidad de Antioquia.
 - * Invited to the 2023 Konstanz School of Collective Behaviour
- Freddy Angarita-Cuesta, MECE Rice University 2023-2024
- Max Bowman, MECE Rice University 2023-2024 (Now at Microsoft)
- William Pan, Undergraduate Researcher ECE Rice University. 2023
- Katelanny Diaz, **SURF Undergraduate Scholar**, CS Rice University, Summer 2023
- Ashwin Aravind, **Mehta Scholars Rice University**, IIT Bombay, Fall 2023
- Annie Xu, **Wagoner Scholar**, STATS Rice University, Summer 2023
- Isabella Estes, **SER scholar**, Undergraduate Researcher CS Rice University. 2022-2023
- William E. Retnaraj, **Visiting Undergraduate Researcher**, IIT Kharagpur, Summer 2023
- Sejal Gupta, **Fondren Fellow**, Undergraduate Researcher CS Rice University. 2022-2023
- Zachary Katz, **Undergraduate Researcher ECE**, Summer 2022 CS Rice University.
- Emily Zhang, **Undergraduate Researcher Google-Rice Program**, Summer 2022 Math Wellesley College.
- Jelena Lazić, **Visiting Researcher Rice-Serbia Program**, Summer 2022 University of Belgrade.
- Johanna Castellanos, **Fulbright Visiting Scholar Fall 2021-Spring 2022**, PhD ECE Universidad Javeriana.
- Ekaterina Trimbach, **Visiting Scholar Spring 2022**, MSc. Student EPFL.
- Eduardo Landin, **Undergraduate Researcher**, Fall 2021 CS Rice University.
- (Yuki) Hanyan Zhang, **Undergraduate Researcher Google-Rice Program**, Summer 2021 CS Florida State University.
- **Student Mentorship & Advocacy**
 - Kai Hung (while Undergraduate at Rice → PhD student at MIT)
 - Gabriel Zook (while Undergraduate at Rice → PhD student at UCSB)
 - Bohan Wu (while Undergraduate at Rice → PhD student at Columbia)
 - Arie Ogranovich (while Undergraduate at Rice → PhD student at UCSB)
 - Yufeng Yang (while Masters at Rice → PhD student at University of Utah)
 - Gilberto Diaz (while Undergraduate at U. de los Andres → PhD student at UCSB)
 - Ferran Arqué (while Undergraduate at Universitat Politècnica de Catalunya)
 - Daniel Ochoa (while Undergraduate at U. de Los Andes → PhD student at UCSD)
 - Eder Baron-Prada (while Undergraduate at U. Nacional → PhD student at ETH)
 - Jingzhao Zhang (while PhD Student at MIT → Assistant Professor at Tsinghua, IIIS)
 - Darina Dvinskikh (while PhD Student at WAIS → Assistant Professor at HSE University)
 - Eduard Gorbunov (while PhD Student at MIPT → Postdoc at Université de Montréal and Mila)
 - Alexander Rogozin (while Master Student at MIPT → Ph.D. student at MIPT)
 - Andres Gómez (while Undergraduate at U. of Antioquia → Ph.D. Goethe University Frankfurt)
 - Carol Bedoya (while Undergraduate at U. of Antioquia → Ph.D. U. of Canterbury)
- **PhD Defense Committee**
 - David Ohlin, Lund University, Department of Automatic Control, 2024
 - Nicolas Zilberstein, Rice University ECE Department, 2024
 - Aysenur Karagoz, Rice University CMOR Department, 2024
 - Dane Grundvig, Rice University CMOR Department, 2024
 - Haoran Li, Rice University, CS Department, 2024
 - Tony Zhang, Rice University, CS Department, 2023
 - Ed Hu, Rice University, CS Department, 2023
 - Adrian Celaya, Rice University CMOR Department, 2024
 - Itzel Olivos Castillo, Rice University, CS Department, 2023
 - John Zito, Rice University, Department of Statistics 2023
 - Jasper Liao, Rice University, CS Department 2023

- Zachary Jansen, Rice University Systems Synthetic and Physical Biology 2023
- Carlos Quintero-Pena Rice University, CS Department, 2023
- Yuxin Tang, Rice University, CS Department, 2023
- Tiancheng Qin, University of Illinois at Urbana-Champaign, ISE Department 2022
- Mohammad Zalbagi Darestani, Rice University, ECE Department, 2022
- Charles Monnoyer de Galland de Carnières, Université Catholique de Louvain, 2022
- Dimitrije Jankov, Rice University CS Department, 2022
- Nathaniel Kroeger, Rice University CAAM Department, 2022
- Eric Antley, Rice University CAAM Department, 2021
- Berkay Turan, University of California Santa Barbara, 2021
- Nestor David Rendon Hurtado, Universidad de Antioquia, Colombia, 2021
- Jhon Jair Quiza Montealegre, Universidad de Antioquia, Colombia, 2021
- Diana Maria Gomez Jaramillo, Universidad de Antioquia, Colombia, 2021
- Wicak Ananduta, UPC Universitat Politècnica de Catalunya, 2019

- **Master Defense Committee**

- Kai Malcolm, Rice University ECE Department, 2024
- Kris Li, Rice University ECE Department, 2023
- Nicolas Zilberstein, Rice University ECE Department, 2023
- Tu Nguyen, Rice University CMOR Department, 2023
- Abhishek Kumar, Rice University ECE Department, 2022
- Madeline Navarro, Rice University ECE Department, 2022
- Maria Isabel Cano Achuri, Universidad de Antioquia, Colombia, 2022
- Sina Alemohammad, Rice University ECE Department, 2022
- Maria Jose Guerrero Muriel, Universidad de Antioquia, Colombia, 2021

INVITED TALKS, POSTERS, AND ABSTRACTS

- **On Graphs with Finite-Time Consensus and Their Use in Gradient Tracking**
 - *Invited Seminar, Department of ECE, Rutgers University, 2023*
 - *Workshop on Dist. Control, Opt. and Learning for Multi-agent Systems, CDC, 2023*
 - *INFORMS JFIG Early-Career Researchers Webinar, 2024*
 - *Invited Seminar, Autonomy Talks ETH, 2024*
 - *Information Theory and Applications Workshop ITA, 2024*
 - *Invited Seminar, Stockholm University, 2024*
 - *Invited Seminar, KTH, 2024*
 - *Invited Seminar, University of Texas at Austin, 2024*
 - *Invited Talk, INFORMS/ALIO/ASOCIO International Conference, 2024*
 - *Invited Talk, MAPI3 Colombian Applied Mathematics Conference, 2024*
 - *Invited Talk, Stochastic Networks Conference, 2024*
- **On First-Order Meta-Reinforcement Learning with Moreau Envelopes**
 - *MOPTA Conference, Advances in Continuous Optimization for Learning, 2023*
- **Decentralized Parametric Inference**
 - *Statistics Colloquium Lecture Series - Department of Statistics, Rice University, 2023*
- **Personalized Asynchronous Federated Learning**
 - *Invited Seminar, 2023 ELLIIT Focus Period, Linköping University, 2023*
 - **KEYNOTE AISTATS LXAI, 2023**
 - *Invited Seminar, Center for Control, Dynamical Systems, and Computation, UCSB, 2023*
 - *EnCORE Session on Distributed Learning Decision Making ITA, 2023*
- **Optimal transport-based segmentation and classification of ECG signals: Preliminary results and challenges**
 - *2022 AI in Health Conference, 2022*
 - *5th Annual Meeting of the SIAM Texas-Louisiana Section*
- **Robust Optimization over Networks Using Distributed Restarting of Accelerated Dynamics**
 - *NeurIPS 2020 LXAI Workshop, Spotlight*
 - *2021 CPS IoT Week workshop on "Computation-Aware Algorithmic Design for Cyber-Physical Systems (CAADCPS 2021)"*
- **Robust Asynchronous and Network-Independent Cooperative Learning**
 - *NeurIPS 2020 LXAI Workshop*
- **A Distributed Cubic-Regularized Newton Method for Smooth Convex Optimization over Networks**
 - *3rd IMA and OR Society Conference on Mathematics of Operational Research, 2021.*
 - *Workshop on "Beyond first-order methods in ML systems" at International Conference on Machine Learning, 2020*
 - *INFORMS Annual Meeting 2020*
- **Hyperfast Second-Order Method for Distributed Convex Optimization**
 - *Workshop on "Beyond first-order methods in ML systems" at International Conference on Machine Learning, 2020*
 - *Yahoo! Research Seminar, 2021.*

- **Near-optimal tensor methods for minimizing the gradient norm of convex functions**
 - Workshop on "Beyond first-order methods in ML systems" at International Conference on Machine Learning, 2020
 - ~~22nd Conference of the International Federation of Operational Research Societies (IFORS 2020)~~. (COVID19)
- **Graph-Theoretic Analysis of Belief System Dynamics under Logic Constraints**
 - ACM Collective Intelligence 2020
- **Towards Scalable Algorithms for Distributed Optimization and Learning**
 - **KEYNOTE** 22nd Mexican International Conference on Artificial Intelligence, 2023.
 - Data-Enabled Science Seminar, Department of Mathematics, University of Houston, 2023.
 - Invited Seminar, Department of Mathematics, Texas State University, 2022.
 - Invited Seminar, Department of ISE, Texas A&M University, 2022.
 - Colombian Conference on Applied and Industrial Mathematics, 2022.
 - Invited Research Seminar, Meta Research, 2022.
 - Summer School 2021, Institucion Universitaria Pascual Bravo, 2021.
 - ZJU-CSE Summer School, Zhejiang University, 2021.
 - CAAM Department, Rice University, 2021.
 - Mechanical Engineering Department, University of Nebraska at Lincoln, 2021.
 - AI LAB Webinar, FPT Software AI Lab, Vietnam, 2021.
 - CEE Graduate Seminar, Rice University, 2021
 - Department of Systems Engineering and Engineering Management, The Chinese University of Hong Kong 2020
 - Department of Mathematics, University of California Davis 2020
 - ECE Department, Carnegie Mellon University 2020
 - ECE Department, Rice University 2020
 - ECE/ISE School, Purdue University 2020
 - ECE Department, GeorgiaTech 2020
 - ECE Department, University of Maryland 2020
 - CSE Department, Pennsylvania State University 2020
 - EECS Department, University of Michigan 2020
 - EE Department, Yale 2020
 - SEMTE School, Arizona State University 2020
 - ECE Department, University of Illinois at Chicago 2020
 - ISE Department, Rensselaer Polytechnic Institute 2020
 - The Graduation Day – 2020 Information Theory and Applications Workshop
- **On the Complexity of Approximating Wasserstein Barycenters**
 - INFORMS Telecommunications and Network Analytics Conference 2020
 - INFORMS Annual Meeting 2019
 - Invited Seminar at Rensselaer Polytechnic Institute 2019
 - 23rd International Symposium on Mathematical Programming ISMP 2018
- **Achieving Acceleration in Distributed Optimization via Direct Discretization of the Heavy-Ball ODE**
 - Modeling and Optimization; Theory and Applications MOPTA 2019
 - International Conference on Continuous Optimization ICCOPT 2019
- **Accelerated Curvature-aided Incremental Aggregated Gradient Method**
 - INFORMS Annual Meeting 2018
 - 23rd International Symposium on Mathematical Programming ISMP 2018
- **Optimal Algorithms for Distributed Optimization**
 - 5th International Conference on Quasilinear Equations, Inverse Problems and their Applications QIPA2019
 - 23rd International Symposium on Mathematical Programming ISMP 2018
 - INFORMS Optimization Society Conference 2018
 - Information Theory and Applications Workshop ITA 2018
- **A Dual Approach for Optimal Algorithms for Distributed Optimization over Networks**
 - ~~SIAM Conference on Optimization 2020~~. (COVID19)
 - ~~INFORMS Optimization Society Conference, 2020~~. (COVID19)
 - INFORMS Annual Meeting, 2018.
 - Midwest Machine Learning Symposium MMLS 2018 **Spotlight Presentation**.
 - 7th Midwest Workshop on Control and Game Theory 2018
 - CSL Student Conference, 2018.
- **Distributed Learning for Cooperative Inference**
 - Science Park Informal Probability Meetings, CWI 2017
 - DCSC Colloquium, Delft University of Technology 2017
 - LCCC Seminar, Lund University 2017
 - SAMSI Workshop on the Interface of Statistics and Optimization WISO 2017
 - SIAM Conference on Optimization 2017
- **Non-asymptotic Rates in Distributed Learning**
 - International Conference on Continuous Optimization ICCOPT 2016
 - SIAM Annual Meeting 2016

- *Stochastic Networks Conference* 2016
- *5th Midwest Workshop in Control and Game Theory* 2016
- *CSL Student Conference* 2016
- *IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems* 2015

LEADERSHIP AND PROFESSIONAL SERVICE

• Faculty Search Committee	Rice Department of Computational and Applied Mathematics	2021
	Rice Department of Statistics	2021
	Rice Data-to-Knowledge Laboratory	2021 – 2022
	Rice ECE Department	2022
• Admissions Committee	PhD Program, Rice ECE Department	2021 – 2023
	Research Experience for Undergraduates, Rice ECE Department	2021 – 2022
	MIT ReaACT Program for Refugees	2018 – 2023
• Research Panels	Two Panels for National Science Foundation (NSF)	2022
	National Research, Development and Innovation Office of Hungary, panel on Mathematics and Computer Science	2019
	C3ai.DTI	2020
	US Army Research Office	2023
• DEI Committee	Rice ECE Department	2021
	NSF BPC Department Delegate	2023
• Representative Member	LIDS-MIT Postdoc Committee	2018 – 2019
• Main Officer	MIT Colombian Association	2018 – 2020
• President	Colombian Student Association at UIUC	2015 – 2016
• Student Representative	Education Committee, Delft Center for Systems and Control, TUDelft.	2011 – 2012
• Student Mentor	Delft Center for Systems and Control, TUDelft	2012
• Jury	LIDS Student Conference: Control, Autonomy, and Decision Making	2012
	Capstone Project Data-to-Knowledge Lab, Rice University	2021 – 2023
	IBB Summer Undergraduate Research Symposium	2023
	Rice Summer Undergraduate Research Symposium	2023

Session Organizer (SO), Organizer (O), Panelist (P) Program Committee (PC) & Session (Co-)Chair (C):

– (O)	NedićFest! A Workshop in Honor of Angelia Nedić	IEEE CDC 2024
– (C)	Reinforcement Learning Session	IEEE CDC 2024
– (C)	Robotics Session	IEEE CCTA, 2024
– (C)	Learning and Intelligence Session	IEEE CCTA, 2024
– (C)	Networks and Telecom Session	ISMP, 2024
– (O)	2024 Rice Workshop on Applied Algorithms for Machine Learning	Rice-AAML, 2024
– (O)	2024 INFORM Optimization Society (IOS) Conference	IOS, 2024
– (SO)	Distributed Optimization Session	IOS, 2024
– (SO)	New Frontiers in Optimization and Data Science Session	IOS, 2024
– (O)	(VC STUDENT AFFAIRS) American Control Conference	ACC, 2025
– (PC)	5th IFAC Workshop on Cyber-Physical Human Systems	CPHS, 2024
– (SO&C)	Complex networks- Analysis, Control and Observability	ECC, 2024
– (SO&C)	Spreading Processes in Complex Networks	ACC, 2024
– (SO)	Modern Applications of Signal and Information Processing	Asilomar Conference 2023
– (P)	International Workshop on AI-Powered Renewable Energy	ION 2023
– (O)	Workshop on Dist. Control, Opt. and Learning for Multi-agent Systems	IEEE CDC, 2023
– (SO&C)	Distributed Optimization and Learning for Networked Systems	IEEE CDC, 2023
– (O)	2023 Texas Colloquium on Distributed Learning	TL;DR 2023
– (PC)	ICML Workshop on Federated Learning and Analytics in Practice	ICML, 2023
– (SO)	Spreading Processes in Complex Networks	ACC, 2023
– (SO&C)	Distributed Optimization and Learning for Networked Systems I, II, III, IV	IEEE CDC, 2022
– (C)	Decentralized Control Session	ACC, 2022
– (C)	Optimization and Control Session	MAPI 2, 2022
– (SO)	Advances on Non-smooth and Non-Convex Optimization	INFORMS Annual Meeting 2022
– (SO)	Theory and Applications of Optimal Transport	Asilomar Conference 2022
– (P)	Combined Colloquium: Diversity, Equity and Inclusion	INFORMS Annual Meeting, 2021
– (O)	Rice-LATAM: Networks, Robots and Machine Learning	Rice University, 2021
– (SO&C)	Recent Advances in High-Order Methods	INFORMS Annual Meeting, 2021

- (SO&C) Distributed Optimization and Learning for Networked Systems I, II, III, IV
 - (C) Optimization II Session
 - (C) Gulf Coast Undergraduate Research Symposium
 - (SO&C) Advances in Distributed Optimization, Estimation and Learning
 - (SO&C) Adversary-Resilient Distributed Machine Learning
 - (PC&C) ICML 2020 LXAI Research Workshop
 - (SO&C) Distributed Optimization and Learning for Networked Systems I, II & III
 - (SO) ~~Convex Optimization and Applications (COVID19)~~
 - (SO) ~~Optimization Methods in Decision and Control (COVID19)~~
 - (SO) ~~Recent Advances in non-Convex/non-Smooth optimization (COVID19)~~
 - (PC) LatinX in AI Research Workshop
 - (C) Optimization Session
 - (SO&C) Large-Scale Distributed Optimization and Decentralized Control I&II
 - (SO&C) Recent Advances in Large-Scale optimization
 - (SO&C) Recent Advances in Distributed Optimization
 - (C) Statistical Learning Session
 - (SO&C) Large-Scale Distributed Optimization and Computation on Graphs
 - (C) Machine Learning and Data Science Session
 - (PC&C) IEEE Colombian Conference on Automatic Control 2017 & 2019 & 2021
 - (SO) Workshop on Distributed Optimization: From consensus to learning
 - (PC) VI Colombian Conference at Harvard/MIT/BU
- IEEE CDC, 2021
 - IEEE CDC, 2021
 - GCURS Rice, 2021
 - SIAM Opt, 2021
 - Asilomar, 2021
 - ICML 2020-2021
 - IEEE CDC, 2020
 - IFORS, 2020
 - IFORS, 2020
 - SIAM Opt, 2020
 - NeurIPS 2019-2020
 - QIPA, 2019
 - IEEE CDC, 2019
 - INFORMS Annual Meeting, 2019
 - ICCOPT, 2019
 - ACC, 2019
 - IEEE CDC, 2018
 - SIAM Annual Meeting 2016
 - CCAC 2017 & 2019 & 2021
 - CCAC 2017
 - Shaping Colombia: 2019

Reviewer

Journals

- Archives of Electrical Engineering (x1)
- ELSEVIER Automatica (x9)
- ELSEVIER Computers & Operation Research (x1)
- ELSEVIER Journal on Applied Soft Computing (x6)
- ELSEVIER Journal of Comp. and Applied Math. (x1)
- ELSEVIER Journal of Computational Science (x1)
- ELSEVIER Journal of the Franklin Institute (x1)
- ELSEVIER Neural Networks (x2)
- ELSEVIER Nonlinear Analysis: Hybrid Systems (x3)
- ELSEVIER Systems & Control Letters (x4)
- IEEE/ACM Transactions on Networking
- IEEE Control Systems Letters (x13)
- IEEE Intelligent Systems (x1)
- IEEE Open Journal of Signal Processing (x2)
- IEEE Open Journal of Control Systems (x1)
- IEEE Signal Processing Letters (x1)
- IEEE Signal Processing Magazine (x1)
- IEEE Trans. on Automatic Control (x50)
- IEEE Trans. on Automation Science and Engineering (x3)
- IEEE Trans. on Control of Network Systems (x20)
- IEEE Trans. on Control Systems Technology (x1)
- IEEE Trans. on Cybernetics (x5)
- IEEE Trans. on Information Theory (x5)
- IEEE Trans. on Signal Processing (x32)
- IEEE Trans. on Signal and Inf. Processing over Networks (x7)
- IEEE Trans. on Network Science and Engineering (x14)
- IET Control Theory & Applications (x3)
- International Journal of Robust and Nonlinear Control (x3)
- Journal of Biomedical and Health Informatics (x1)
- Journal of Inverse and Ill-Posed Problems (x2)
- Journal of Machine Learning Research (x9)
- Journal of Global Optimization (x1)
- JOTA Journal of Optimization Theory & Applications (x2)
- Mathematical Programming (x1)
- OPRE Operations Research (x2)
- OPTL Optimization Letters (x1)
- GOMS Optimization Methods and Software (x1)
- SIOPT SIAM Journal on Optimization (x1)
- SICON SIAM Journal on Control and Optimization (x2)
- SSy Stochastic Systems (x2)

Conferences

- ACM/IEEE International Conference on Cyber-Physical Systems ICCPS 2017, 2020.
- American Control Conference ACC 2014-2023.
- Conference on Learning Theory COLT 2019-2021.
- Conference on Neural Information Processing Systems NeurIPS 2019-2022.
- Colombian Conference on Bioacoustics and Ecoacoustics 2022.
- European Control Conference ECC 2014-2018.
- IEEE 15th International Conference on Automation Science and Engineering CASE 2019.
- IEEE Colombian Control Conference 2015, 2017, 2019, 2021.
- IEEE Control and Decision Conference CDC 2011-2022.
- IEEE Information Theory Workshop 2019.
- IEEE International Conference on Acoustics, Speech, and Signal Processing, ICASSP 2020
- IEEE Multi-conference on Systems and Control 2016.
- Indian Control Conference 2022.
- IFAC Workshop on Distributed Estimation and Control in Networked Systems NecSys 2016, 2019, 2022
- International Conference on Artificial Intelligence and Statistics AISTATS 2021-2022.
- International Conference on Machine Learning ICML 2016, 2020-2022.
- International Conference on Learning Representations ICLR 2021-2023.
- International Symposium on Stabilization, Safety, and Security of Distributed Systems SSS 2016.
- Learning for Dynamics and Control L4DC, 2020-2021.
- NeurIPS 2021 Workshop on Federated Learning: New Challenges on Privacy, Fairness, Robustness, Personalization and Data Ownership

Proposal Evaluation Panels

- Editorial Board of the Journal of Machine Learning Research
- Springer Book Series Reviewer.
- National Science Foundation x2.
- National Research, Development and Innovation Office of Hungary, panel on Mathematics and Computer Science.
- C3ai.DTI COVID-19 Proposal Reviewer.

PROFESSIONAL ORGANIZATION MEMBERSHIPS

- Institute for Operations Research and the Management Sciences INFORMS
- Institute of Electrical and Electronics Engineers IEEE
- Society for Industrial and Applied Mathematics SIAM
- Association for Computing Machinery ACM

ADDITIONAL EXPERIENCE

Spring 2024	<i>Visiting Scholar</i> , ICERM Program on Interacting Particle Systems: Analysis, Control, Learning and Computation, Brown University
Fall 2023	<i>Visiting Scholar</i> , ELLIIT Focus Period on Network Dynamics and Control, Linköping University
Spring 2023	<i>Visiting Scholar</i> , ICERM Program on Optimal Transport and Data Science, Brown University
Summer 2017	<i>Visiting Scholar</i> , LCCC Focus Period on Large-Scale and Distributed Optimization, Lund University.
Spring 2017	<i>Visiting Scholar</i> , Arizona State University.
August 2013	<i>Attendee</i> , IMSE Summer School on Multi-Agent Networked Systems, University of Illinois Urbana-Champaign.
July 2013	<i>Attendee</i> , 5th HYCON2 PhD School on Control of Networked and Large-Scale Systems.
Summer 2011	<i>Visiting Scholar</i> , Laboratoire d'Analyse et d'Architecture des Systemes (LAAS), France.
Summer 2010	<i>Visiting Scholar</i> , InfoLab, University of Delaware, Delaware, USA.